Name: ________________________, Roll No.: ____________

Total marks

CSE 350/550: Network Security, Quiz 2: April 7, 2026

Instructions:

- There are 3 questions, each 10 marks, for a total of 30 marks. Total time available 30 minutes.
- Provide answers only in the space provided. You may do your rough work on a separate blank sheet.
- You may use a calculator, but nothing else. **Pl. keep your phones/laptops, books etc. away from you**.
- Do not use unfair means, else action will be taken. Do not use material such as books, notes, computer, smart phone, etc. And do not share information with other students.

Q1

1. Bob has decided to send two messages, M1, M2 to Alice suitably encrypted using ElGamal Cryptosystem (discussed in class). To do so, Alice and Bob plan to generate shared keys using Diffie-Hellman method but using the concept of ONE-TIME-KEYS. The Diffie-Hellman method is characterized by **prime q=19, with primitive root α = 10**.

Consequently, Alice selects "private key" XA1 from its list of **random numbers (5, 7, 10, ...),** while Bob selects "private keys" XB1, XB2 from its list of **random numbers (12, 3, 7, ...).**
Clearly Alice computes the corresponding "public key" YA1 and shares it with Bob. On his part, Bob computes the corresponding "public keys" YB1, YB2 and shares them with Alice, together with cyphertexts C1, C2 respectively for **messages M1=6, M2=8**. Note, Bob obtains C1 and C2 only after he computes shared keys, KB1, KB2 and encrypts the messages M1 and M2 using ElGamal cryptosystem.
Alice, computes the shared key KA1, KA2, etc. and de-ciphers C1, C2, etc. to recover messages M1, M2.

**Show all your calculations below:**

**At Alice's end:**

XA1 = ___5 ___; YA1 = ___ 3____;

Alice sends to Bob = (___3___)

Calculations concerning **message M1 = 6**

**At Bob's end:**

XB1 = ___12___; YB1 = ___7___;

KB1 = ____11__;

M1 = ___6__; C1 = ___9___;

BOB sends to Alice = (___7___, ___9___)

**At Alice's end:**

KA1 = __11____; inverse(KA1) = ___7__;

Computed M1 = ___6___;

Calculations concerning **message M2 = 8**

**At Bob's end:**

XB2 = ___3___; YB2 = ___12___;

KB2 = ___8___;

M2 = ___8___; C2 = ___7____;

BOB sends to Alice = (____12____, ____7__)

**At Alice's end:**

KA2 = __8____; inverse(KA2) = ___12__;

Computed M2 = ___8__;

**Give 1/2 points for each value highlighted by yellow → total of 8 points. Give another 2 point if content of "one-time-keys" is understood.**

Some students may have picked different values for XA1, XB1, and XB2. If such cases the TA will have to compute all the numbers, viz. YA1, etc., KA1, etc., C1, etc., and thereby recover M1, etc.

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Q2

2. Take a look at table below that describes a message together with the hash value. Here characters x_0, x_1, … x_6 are characters that form the message. The 8-bits of any character x_j are shown as:
xj7, xj6, …, xj0, **where xj7 is the (EVEN) parity bit obtained by adding bits row-wise as follows**. That is:
x07 + x06 + … + x00 = 0; x17 + x16 + … + x10 = 0; …, x67 + x66 + … + x60 = 0
A character x_7 is appended. It forms part of the hash value. It is obtained by adding bits column-wise as follows:
x70 + x60 + … + x00 = 0; x71 + x61 + … + x01 = 0; …; x77 + x67 + … + x07 = 0

| Message characters | **(parity) bit_7** | **bit_6** | **bit_5** | **bit_4** | **bit_3** | **bit_2** | **Bit_1** | **bit_0** |
|---|---|---|---|---|---|---|---|---|
| x_0 | x07 | x06 | x05 | x04 | x03 | x02 | x01 | x00 |
| x_1 | x17 | x16 | x15 | x14 | x13 | x12 | x11 | x10 |
| x_2 | x27 | x26 | x25 | x24 | x23 | x22 | x21 | x20 |
| x_3 | x37 | x36 | x35 | x34 | x33 | x32 | x31 | x30 |
| x_4 | x47 | x46 | x45 | x44 | x43 | x42 | x41 | x40 |
| x_5 | x57 | x56 | x55 | x54 | x53 | x52 | x51 | x50 |
| x_6 | x67 | x66 | x65 | x64 | x63 | x62 | x61 | x60 |
| (parity) x_7 | x77 | x76 | x75 | x74 | x73 | x72 | x71 | x70 |

To summarize, the sum of all bits in a given row is 0. And the sum of all bits in a given column is 0. The hash value consists of all the shaded bits from the table, viz. [x77, x67, x57, x47, x37, x27, x17, x07], and [x77, x76, x75, x74, x73, x72, x71, x70]. Use following message if you need to work with an example.

| Message characters | **(parity) bit_7** | **bit_6** | **bit_5** | **bit_4** | **bit_3** | **bit_2** | **Bit_1** | **bit_0** |
|---|---|---|---|---|---|---|---|---|
| x_0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 |
| x_1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 |
| x_2 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 |
| x_3 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| x_4 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 0 |
| x_5 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 |
| x_6 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 1 |
| (parity) x_7 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 |

**Give 3 points for correct answer for parts a. and b., and 4 points for part**

Questions:

a. **Is this hash function "pre-image resistant"? ~~Yes~~/No**. Explain with an example.
**ANS**: For a given h=H(x), it is easy to compute an x, such that H(x)=h. For example, let h be the same as in the above example. Clearly one choice of x is the same as the x in above example, but with characters x_0 and x_2 interchanged. This will also result in H(x) = h.

b. **Is this hash function "second pre-image resistant"? ~~Yes~~/No**. Explain with an example.
**ANS**: For a given x, it is easy to compute a y, y≠x, such that H(y)=H(x). For example, let x be the same as in the above example. Clearly a y constructed from above example by interchanging characters x_0 and x_2 will result in H(x)=H(y).

c. **Is this hash function "collision resistant"? ~~Yes~~/No**. Explain.
**ANS**: Can we easily find an x and a y such that H(x)=H(y), x≠y. Yes, we can. For example, let x be the same as one given above, and y be the same as the one given above BUT with characters x_0 and x_2 interchanged. Clearly x and y, x≠y, so constructed result in H(x)=H(y).

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Q3

3. A portion of **Kerberos v4** protocol is given below.

(1) $\mathbf{C \rightarrow AS}$ $\quad ID_c \| ID_{tgs} \| TS_1$

(2) $\mathbf{AS \rightarrow C}$ $\quad E(K_c, [K_{c,tgs} \| ID_{tgs} \| TS_2 \| Lifetime_2 \| Ticket_{tgs}])$

$$Ticket_{tgs} = E(K_{tgs}, [K_{c,tgs} \| ID_C \| AD_C \| ID_{tgs} \| TS_2 \| Lifetime_2])$$

(a) Authentication Service Exchange to obtain ticket-granting ticket

(3) $\mathbf{C \rightarrow TGS}$ $\quad ID_v \| Ticket_{tgs} \| Authenticator_c$

(4) $\mathbf{TGS \rightarrow C}$ $\quad E(K_{c,tgs}, [K_{c,v} \| ID_v \| TS_4 \| Ticket_v])$

$$Ticket_{tgs} = E(K_{tgs}, [K_{c,tgs} \| ID_C \| AD_C \| ID_{tgs} \| TS_2 \| Lifetime_2])$$

$$Ticket_v = E(K_v, [K_{c,v} \| ID_C \| AD_C \| ID_v \| TS_4 \| Lifetime_4])$$

$$Authenticator_c = E(K_{c,tgs}, [ID_C \| AD_C \| TS_3])$$

(b) Ticket-Granting Service Exchange to obtain service-granting ticket

(5) $\mathbf{C \rightarrow V}$ $\quad Ticket_v \| Authenticator_c$

(6) $\mathbf{V \rightarrow C}$ $\quad E(K_{c,v}, [TS_5 + 1])$ (for mutual authentication)

$$Ticket_v = E(K_v, [K_{c,v} \| ID_C \| AD_C \| ID_v \| TS_4 \| Lifetime_4])$$

$$Authenticator_c = E(K_{c,v}, [ID_C \| AD_C \| TS_5])$$

**Briefly, explain your answers**.

A. Contents of message (2) are encrypted using key $K_C$. What purpose does that serve?
2 points
**ANS**: This allows AS to authenticate client C, or in other words to ensure that only client C is able to decrypt message (2) and thereby extract Ticket_tgs.

B. Which message, and what content in it, allows the Ticket-granting server TGS to authenticate Client, C?
2 points
**ANS**: TGS is able to authenticate client C once it successfully decrypts Authenticator_C in message (3) using the (temporary) key K_(C,tgs) which it extracted securely from Ticket_TGS generatd by AS, but received from client C.

C. Which message, and what content in it, allows the client to authenticate the Ticket-granting server TGS?
2 points
**ANS**: Client C is able to authenticate TGS once it successfully decrypts message (4) using its (temporary) key K_(C,tgs) extracted securely from message 2 generated and sent earlier by AS.

D. What purpose does the key $K_{C,V}$ in message 4 serve?
2 points
**ANS**: It allows the server V to authenticate client C as also allow client C to authenticate sever V.

E. Based on what message and its content, is server V convinced that $Ticket_V$ was generated by TGS?
2 points
**ANS**: Ticket_V was generated by TGS by encrypting it with key K_V that was earlier established between TGS and V. Thus, when Ticket_V, earlier received by server V in message (5), is successfully decrypted by server V it can be sure that Ticket_V was indeed generated by TGS.